

Import Libraries.....

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Data.....

```
In [8]: transactions = pd.read_excel("QVI_transaction_data.xlsx")
customers = pd.read_csv("QVI_purchase_behaviour.csv")
```

```
In [9]: print(transactions.shape)
print(customers.shape)

transactions.head()
customers.head()
```

(264836, 8)

(72637, 3)

```
Out[9]:
```

	LYLTY_CARD_NBR	LIFESTAGE	PREMIUM_CUSTOMER
0	1000	YOUNG SINGLES/COUPLES	Premium
1	1002	YOUNG SINGLES/COUPLES	Mainstream
2	1003	YOUNG FAMILIES	Budget
3	1004	OLDER SINGLES/COUPLES	Mainstream
4	1005	MIDAGE SINGLES/COUPLES	Mainstream

basics data checks...

```
In [10]: transactions.info()
customers.info()

transactions.describe()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 264836 entries, 0 to 264835
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   DATE                   264836 non-null int64
1   STORE_NBR              264836 non-null int64
2   LYLTY_CARD_NBR        264836 non-null int64
3   TXN_ID                 264836 non-null int64
4   PROD_NBR               264836 non-null int64
5   PROD_NAME              264836 non-null object
6   PROD_QTY               264836 non-null int64
7   TOT_SALES              264836 non-null float64
dtypes: float64(1), int64(6), object(1)
memory usage: 16.2+ MB
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 72637 entries, 0 to 72636
Data columns (total 3 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LYLTY_CARD_NBR        72637 non-null int64
1   LIFESTAGE              72637 non-null object
2   PREMIUM_CUSTOMER      72637 non-null object
dtypes: int64(1), object(2)
memory usage: 1.7+ MB

```

Out[10]:

	DATE	STORE_NBR	LYLTY_CARD_NBR	TXN_ID	PROD_NBR	
count	264836.000000	264836.000000	2.648360e+05	2.648360e+05	264836.000000	26
mean	43464.036260	135.08011	1.355495e+05	1.351583e+05	56.583157	
std	105.389282	76.78418	8.057998e+04	7.813303e+04	32.826638	
min	43282.000000	1.00000	1.000000e+03	1.000000e+00	1.000000	
25%	43373.000000	70.00000	7.002100e+04	6.760150e+04	28.000000	
50%	43464.000000	130.00000	1.303575e+05	1.351375e+05	56.000000	
75%	43555.000000	203.00000	2.030942e+05	2.027012e+05	85.000000	
max	43646.000000	272.00000	2.373711e+06	2.415841e+06	114.000000	



```

In [11]: transactions . isnull().sum()
customers.isnull().sum()

```

```

Out[11]: LYLTY_CARD_NBR      0
LIFESTAGE                  0
PREMIUM_CUSTOMER          0
dtype: int64

```

```

In [12]: transactions['DATE'] = pd.to_datetime('1899-12-30') + pd.to_timedelta(transaction

```

```

In [13]: transactions = transactions[transactions['PROD_NAME'].str.contains("Salsa", case=

```

```

In [15]: transactions['PACK_SIZE'] = transactions['PROD_NAME'].str.extract('(\d+)').astype

```

Extract brand name

```
In [16]: transactions['BRAND'] = transactions['PROD_NAME'].str.split().str[0]
```

```
In [17]: transactions['BRAND'] = transactions['BRAND'].replace({
    "Red": "RRD",
    "Smith": "Smiths",
    "Dorito": "Doritos",
    "Grain": "GrnWves"
})
```

Merge customer data

```
In [18]: df = pd.merge(transactions, customers, on="LYLTY_CARD_NBR")
```

```
In [19]: df.head()
df.shape
```

```
Out[19]: (18094, 12)
```

Calculate total sales

```
In [20]: total_sales = df['TOT_SALES'].sum()
print("Total Sales:", total_sales)
```

Total Sales: 127937.3

Sales by customer lifestage

```
In [21]: lifestage_sales = df.groupby("LIFESTAGE")["TOT_SALES"].sum().sort_values(ascending=True)
print(lifestage_sales)
```

```
LIFESTAGE
OLDER SINGLES/COUPLES    26407.1
RETIREEES                24089.0
OLDER FAMILIES           23947.3
YOUNG FAMILIES           21532.2
YOUNG SINGLES/COUPLES    16648.7
MIDAGE SINGLES/COUPLES   12227.5
NEW FAMILIES              3085.5
Name: TOT_SALES, dtype: float64
```

Sales by premium customer

```
In [22]: premium_sales = df.groupby("PREMIUM_CUSTOMER")["TOT_SALES"].sum()
print(premium_sales)
```

```
PREMIUM_CUSTOMER
Budget          44804.7
Mainstream      49879.1
Premium         33253.5
Name: TOT_SALES, dtype: float64
```

Number of transactions per segment

```
In [23]: transactions_segment = df.groupby("LIFESTAGE")["TXN_ID"].count()
print(transactions_segment)
```

```
LIFESTAGE
MIDAGE SINGLES/COUPLES    1712
NEW FAMILIES              422
OLDER FAMILIES           3436
OLDER SINGLES/COUPLES    3686
RETIREEES                3332
YOUNG FAMILIES           3098
YOUNG SINGLES/COUPLES    2408
Name: TXN_ID, dtype: int64
```

Average spending per customer

```
In [24]: avg_spend = df.groupby("LYLTY_CARD_NBR")["TOT_SALES"].mean()
print(avg_spend.head())
```

```
LYLTY_CARD_NBR
1010    5.1
1011    5.1
1028    2.7
1057    5.1
1065    2.7
Name: TOT_SALES, dtype: float64
```

Brand performance

```
In [25]: brand_sales = df.groupby("BRAND")["TOT_SALES"].sum().sort_values(ascending=False)
print(brand_sales)
```

```
BRAND
Old      90785.1
Doritos  14261.0
Woolworths  8284.5
RRD      7438.5
Smiths   7168.2
Name: TOT_SALES, dtype: float64
```

Package size popularity

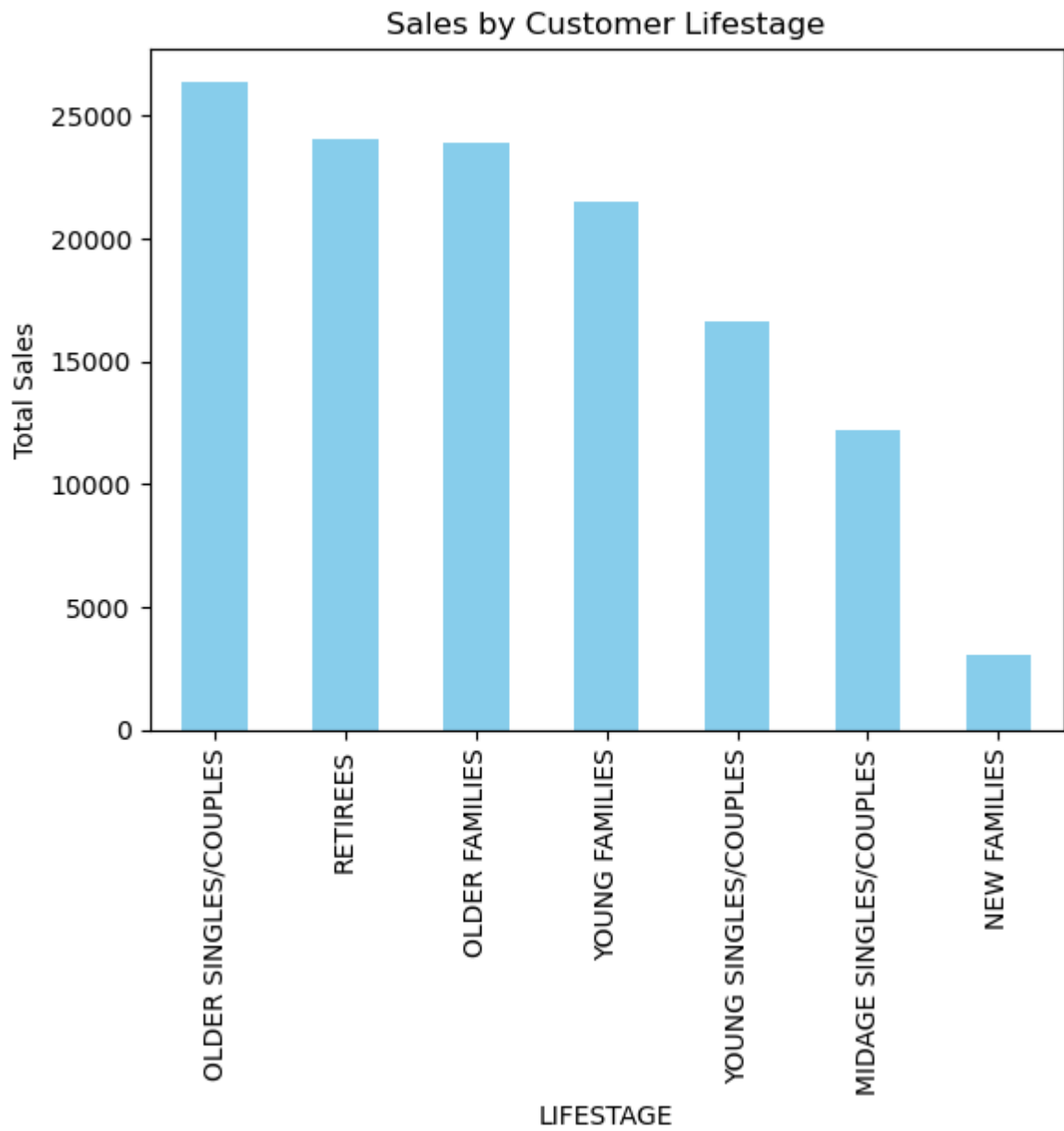
```
In [26]: pack_sales = df.groupby("PACK_SIZE")["TOT_SALES"].sum().sort_values(ascending=False)
print(pack_sales)
```

```
PACK_SIZE
300.0    113330.6
150.0    14606.7
Name: TOT_SALES, dtype: float64
```

Visualization ...

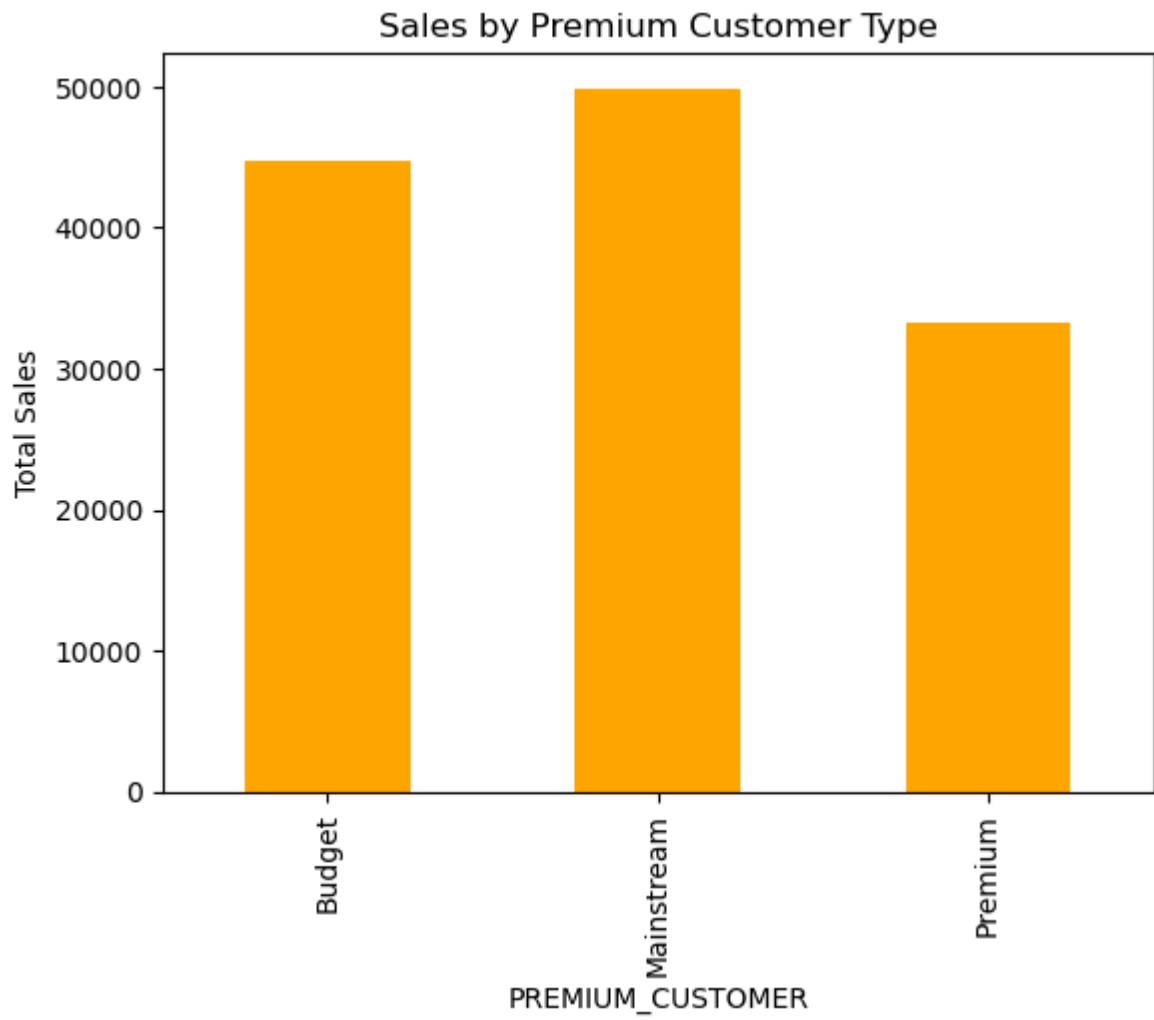
1.Sales by lifestage

```
In [27]: lifestage_sales.plot(kind="bar", color="skyblue")
plt.title("Sales by Customer Lifestage")
plt.ylabel("Total Sales")
plt.show()
```



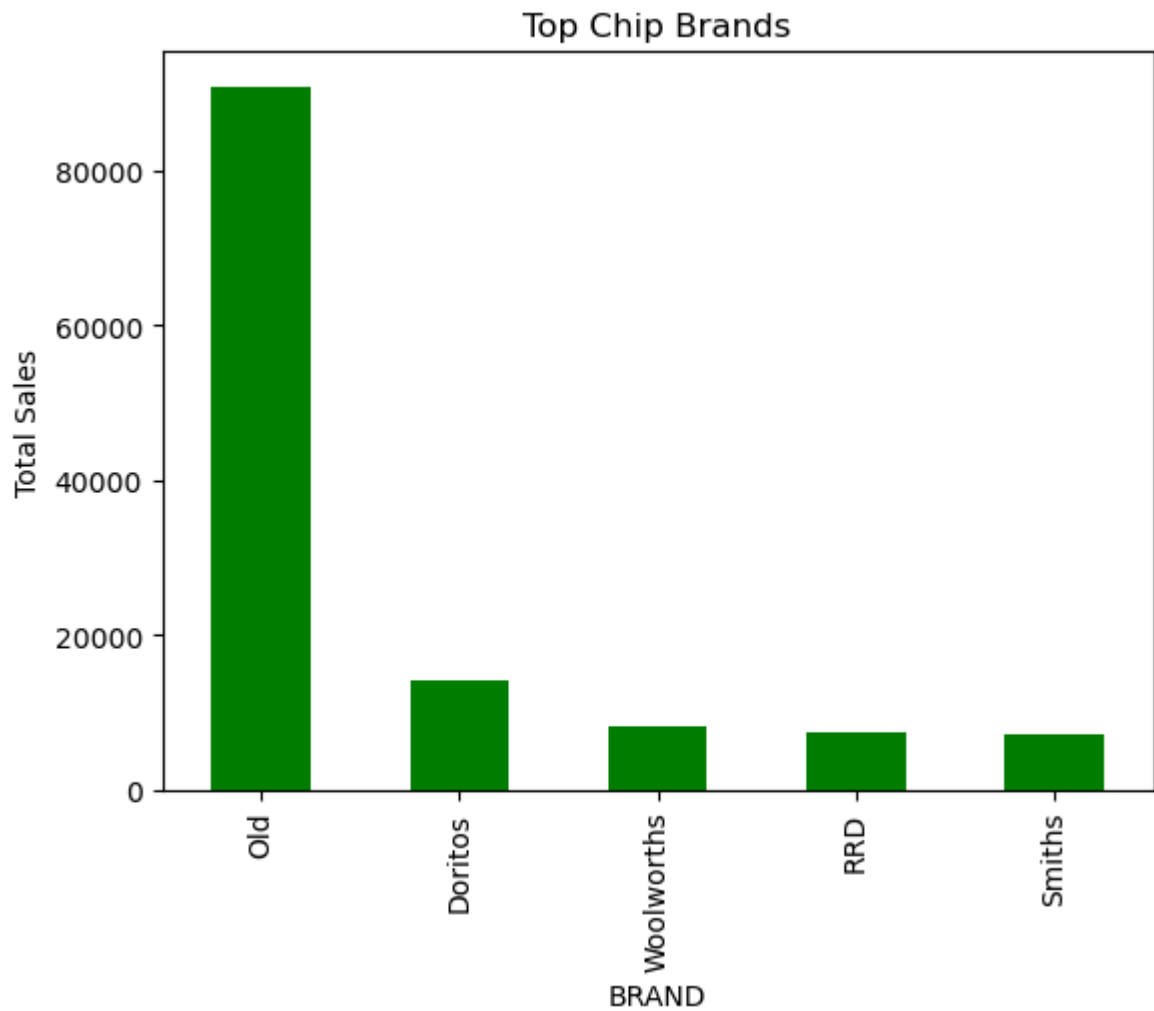
2.Premium vS budget customers....

```
In [28]: premium_sales.plot(kind="bar", color="orange")
plt.title("Sales by Premium Customer Type")
plt.ylabel("Total Sales")
plt.show()
```



3.TOP CHIP BRANDS....

```
In [29]: brand_sales.head(10).plot(kind="bar", color="green")
plt.title("Top Chip Brands")
plt.ylabel("Total Sales")
plt.show()
```



4.PACK SIZE PREFERENCE.....

```
In [30]: pack_sales.head(10).plot(kind="bar", color="purple")
plt.title("Most Popular Pack Sizes")
plt.ylabel("Total Sales")
plt.show()
```

